

UF AI DAYS · OCT. 27, 2026

AI Research, Teaching, and Learning Day

# Restoring the Milky Way

## Scaling AI-assisted light pollution planning across North Central Florida

**Jay Rosen** · jayrosen@ufl.edu
Research Software Engineer III, University of Florida College of Education

Photo: Milky Way and urban skyglow over Paynes Prairie, 2022. Jay Rosen

# About me: from light art to light pollution

- **Light art, 15 years**: light, mirrors, and increasingly technological media
- **Before UF**: founder and CEO, Space Age Mining Corp. (HPC)
- **At UF**: Research Software Engineer III, College of Education
- **Astrophotography since 2020**: photos on state park websites
- **NASA Space Apps**: 2× Kennedy Space Center winner, global nominee, judge
- **Outside UF**: Board of Directors, Alachua Astronomy Club
- Civic service on **City of Gainesville** and **Alachua County** boards



Full-spectrum self-portrait at the Chiefland Astronomy Village, April 2024. Photo: Jay Rosen

THE PROBLEM

# What is light pollution?

**Light pollution**: excessive, misdirected or obtrusive **artificial light at night** (ALAN)

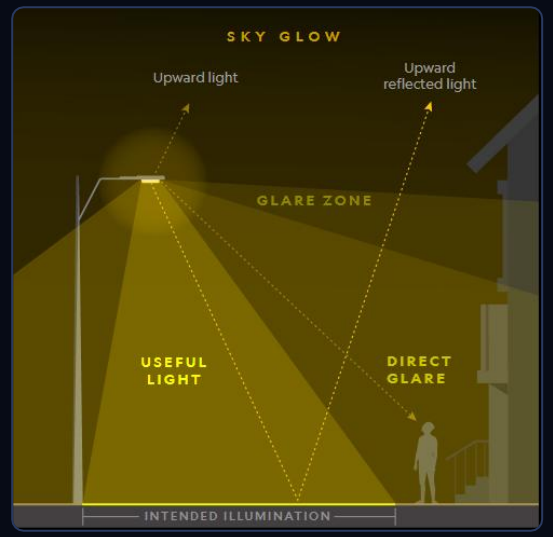



Image: Leeds Astronomical Society

- **Skyglow**: the dome over a city that washes out the stars
- **Light trespass**: spill onto neighbors and habitat
- **Glare**: brightness that hurts visibility
- **Excessive brightness**: more light than the task needs

## Why it matters

- **Wildlife**, as at Paynes Prairie: insects drawn to their deaths, frogs calling less, night-migrating birds thrown off course, bats losing dark habitat
- **People**: blue-rich light at night suppresses melatonin and disrupts sleep
- **Energy**: light sent into the sky is paid for and wasted
- **The night sky**: skyglow hides the Milky Way, even at UF's observatories



Fireflies. Photo: UF IFAS



Bats at dusk. Photo: UF IFAS

Sources: Rosen (2026), Preserving natural skies in Alachua County (EPAC presentation)

THE PROBLEM

# Skyglow travels: one city's lights wash out a whole region



**Paynes Prairie · Bortle 4, Gainesville dome.** Gainesville's skyglow and Comet NEOWISE from Paynes Prairie, July 2020. Photo: Jay Rosen



**Stephen C. Foster State Park, GA · Bortle 2.** International Dark Sky Park, October 2020. Photo: Jay Rosen

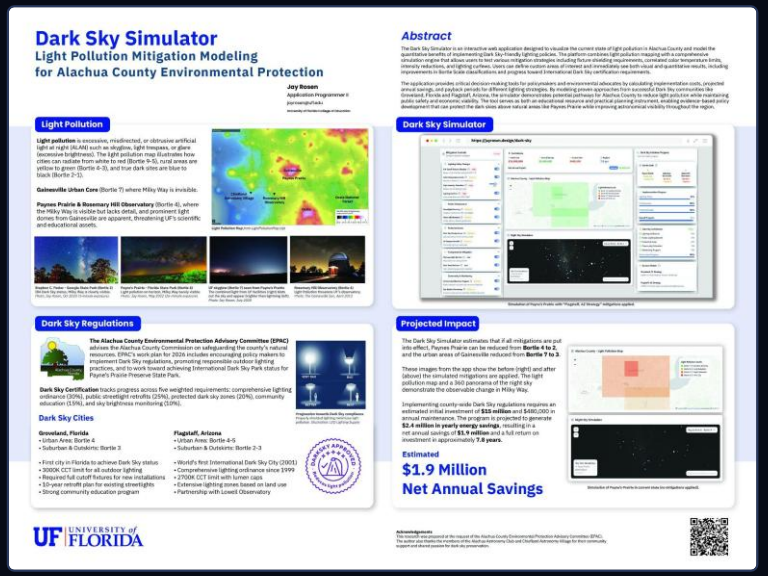
- Light scattered by air and haze brightens the sky **tens of km** away
- Harms wildlife, sleep and health, and wastes energy
- Threatens UF's dark sites: **Rosemary Hill Observatory** and **Chiefland Astronomy Village**
- **Preventable**: shield, aim, warmer color, less brightness, dim late at night

Sources: Garstang (1986), Model for artificial night-sky illumination · Bortle (2001), Introducing the Bortle dark-sky scale · DarkSky International, International Dark Sky Places program guidelines · Rosen (2026), Preserving natural skies in Alachua County (EPAC presentation)

BACKSTORY

# From astrophotography to county policy

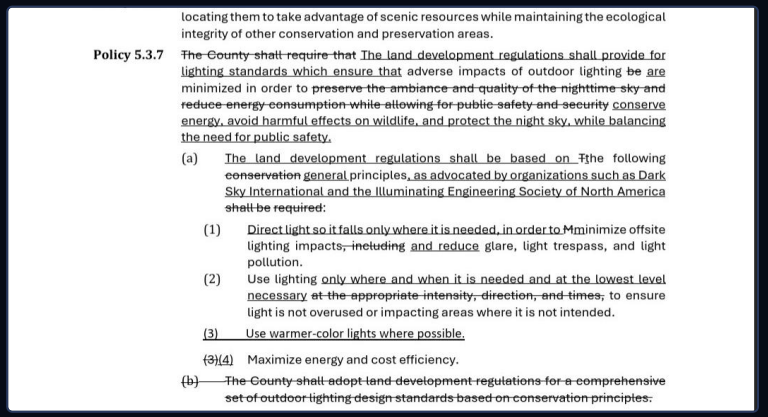
- **2020–25** Photographing the night sky under Gainesville's light dome
- **Sep 2025** Public comment at Alachua County's advisory committee (EPAC)
- **Oct 2025** v1 simulator, built in a month: mitigations' economic and night-sky impact; debuted at UF Astraeus Space Research Day
- **Jan 2026** Presented the research and v1 simulator to EPAC; unanimous vote to pursue dark-sky policy
- **Feb–May 2026** Policy 5.3.7 amendment heard and sent to the state
- **Apr 2026** Dark Sky Week talks: Dr. Lada's UF class asks about a next observatory; the Alachua Astronomy Club asks about Levy
- **Sep 22, 2026** County Commission **adopts** it, unanimously
- **Now** Expanding into **Levy County**; screening sites for UF's next observatory




v1 poster, UF Astraeus Space Research Day, Oct 2025




UF College of the Arts, Mar 2026



Policy 5.3.7 locating them to take advantage of scenic resources while maintaining the ecological integrity of other conservation and preservation areas.

**Policy 5.3.7** ~~The County shall require that~~ The land development regulations shall provide for lighting standards which ensure that adverse impacts of outdoor lighting ~~be~~ are minimized in order to ~~preserve the ambiance and quality of the nighttime sky and reduce energy consumption while allowing for public safety and security~~ conserve energy, avoid harmful effects on wildlife, and protect the night sky, while balancing the need for public safety.

(a) The land development regulations shall be based on ~~t~~The following ~~conservation~~ general principles, as advocated by organizations such as Dark Sky International and the Illuminating Engineering Society of North America ~~shall be required~~:

(1) Direct light so it falls only where it is needed, in order to ~~M~~minimize offsite lighting impacts~~, including~~ and reduce glare, light trespass, and light pollution.

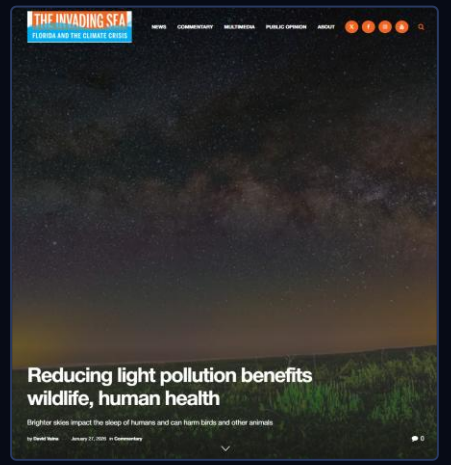
(2) Use lighting only where and when it is needed and at the lowest level necessary ~~at the appropriate intensity, direction, and times,~~ to ensure light is not overused or impacting areas where it is not intended.

(3) Use warmer-color lights where possible.

~~(3)~~(4) Maximize energy and cost efficiency.

~~(b) The County shall adopt land development regulations for a comprehensive set of outdoor lighting design standards based on conservation principles.~~

Policy 5.3.7 language, Feb 2026



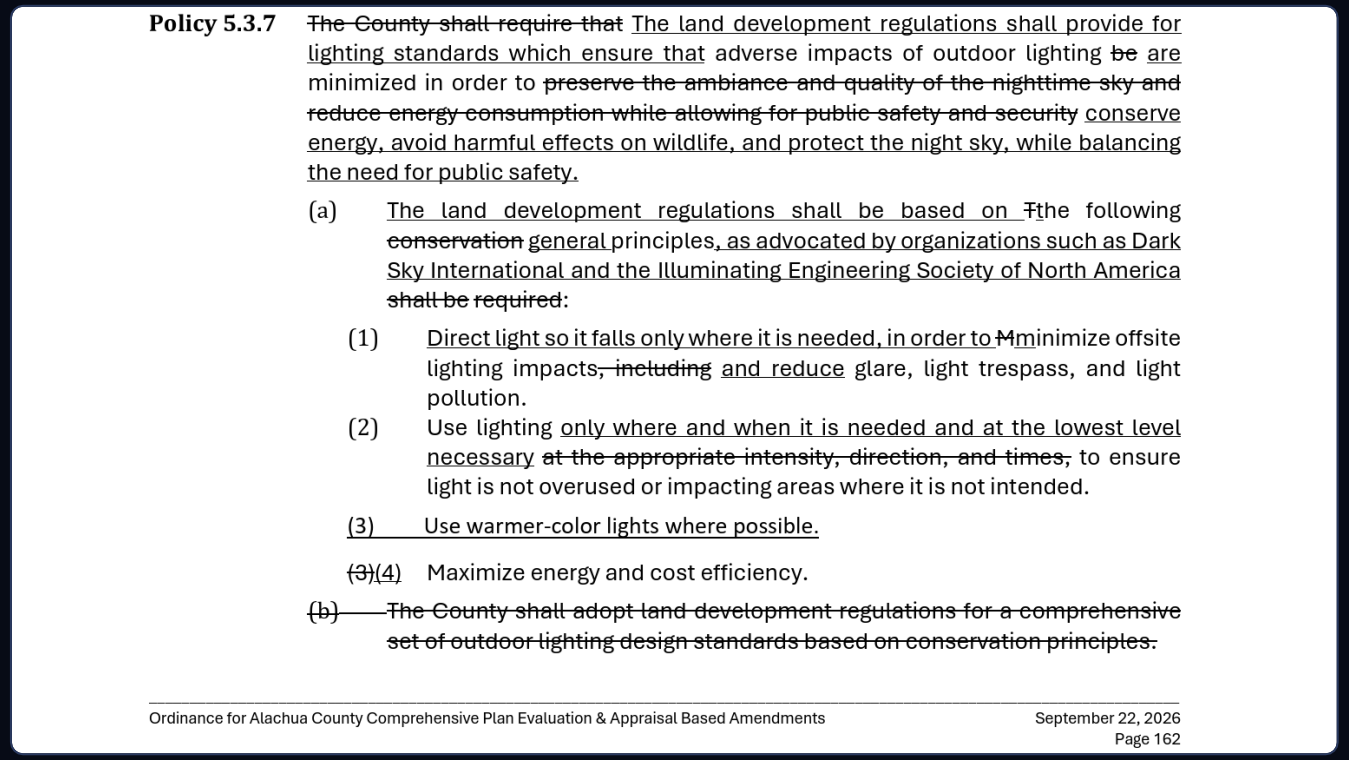

The Invading Sea, Jan 2026

Sources: Rosen (2025), Dark Sky Simulator (v1) · Rosen (2026), Preserving natural skies in Alachua County (EPAC presentation) · Alachua County (2026), Comprehensive Plan amendments, Conservation and Open Space Element, Policy 5.3.7 · Smith (2026), Restoring the Milky Way: Arts alumnus develops app to reduce light pollution

BACKSTORY

# September 2026: dark-sky principles adopted

- Alachua County's Comprehensive Plan update, **Policy 5.3.7** (Conservation and Open Space Element)
- Lighting standards must **conserve energy, avoid harm to wildlife and protect the night sky**, balanced with public safety
- Based on principles from **Dark Sky International** and the **Illuminating Engineering Society**
- Light only where and when needed, at the lowest level; **warmer colors**; energy and cost efficiency
- Each one is a control in the simulator: shielding, dimming, the 3000 K cap, costs
- Next: the land development code; in effect once the state finds it in compliance



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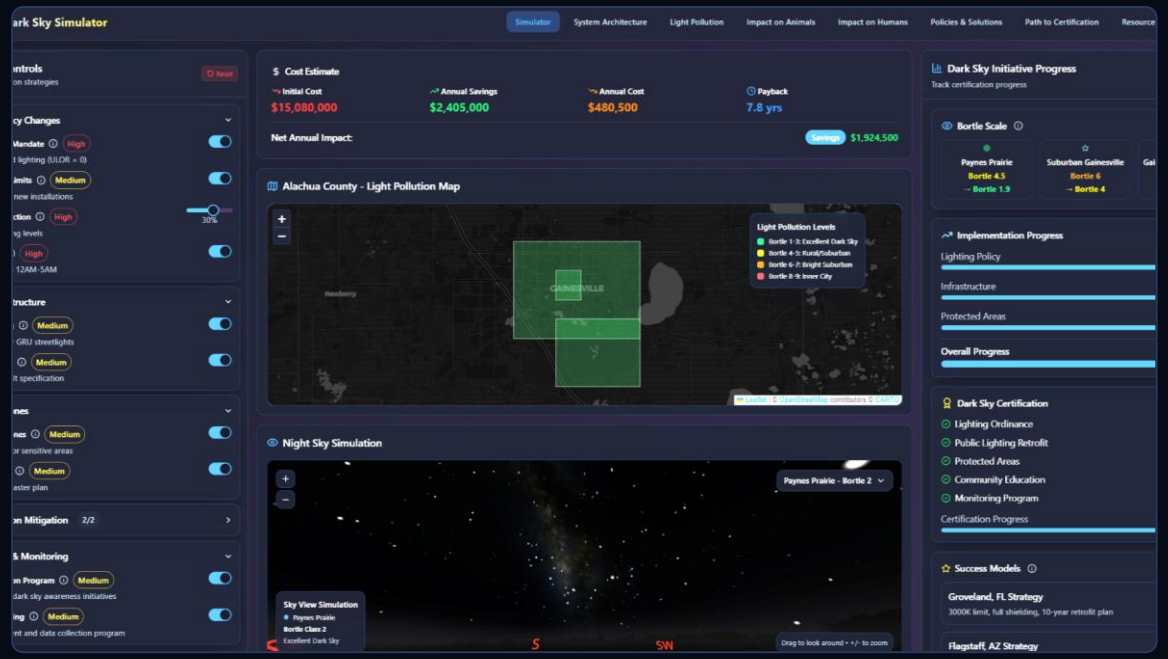
Ordinance for Alachua County Comprehensive Plan Evaluation & Appraisal Based Amendments — September 22, 2026 Page 162

Policy 5.3.7 as adopted: underlined text added, struck text removed. Alachua County ordinance amending the Comprehensive Plan (Evaluation & Appraisal amendments), September 2026, p. 162.

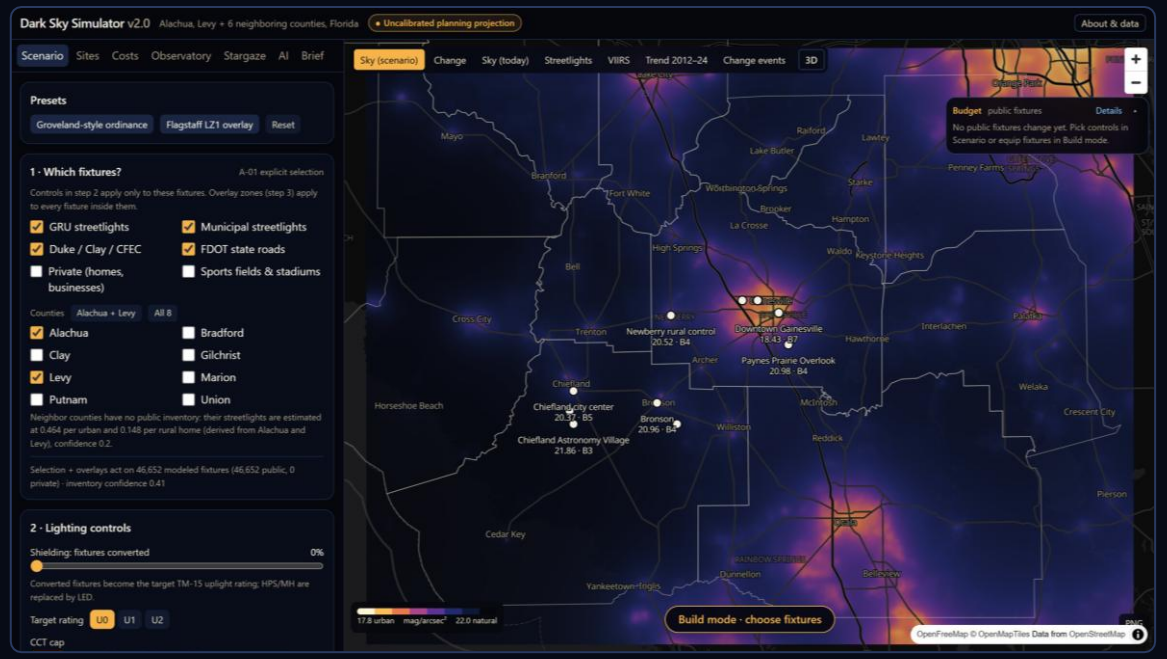
Sources: Alachua County (2026), Comprehensive Plan amendments, Conservation and Open Space Element, Policy 5.3.7 · DarkSky International & IES (2020), Five Principles for Responsible Outdoor Lighting

WHY A VERSION 2

# v1 told a story. v2 has to answer “How do you know?”




v1 · 2025




v2 · 2026

- Hand-tuned multipliers, built in a month
- Persuasive headline numbers (“$1.9M a year saved”) that nobody could audit

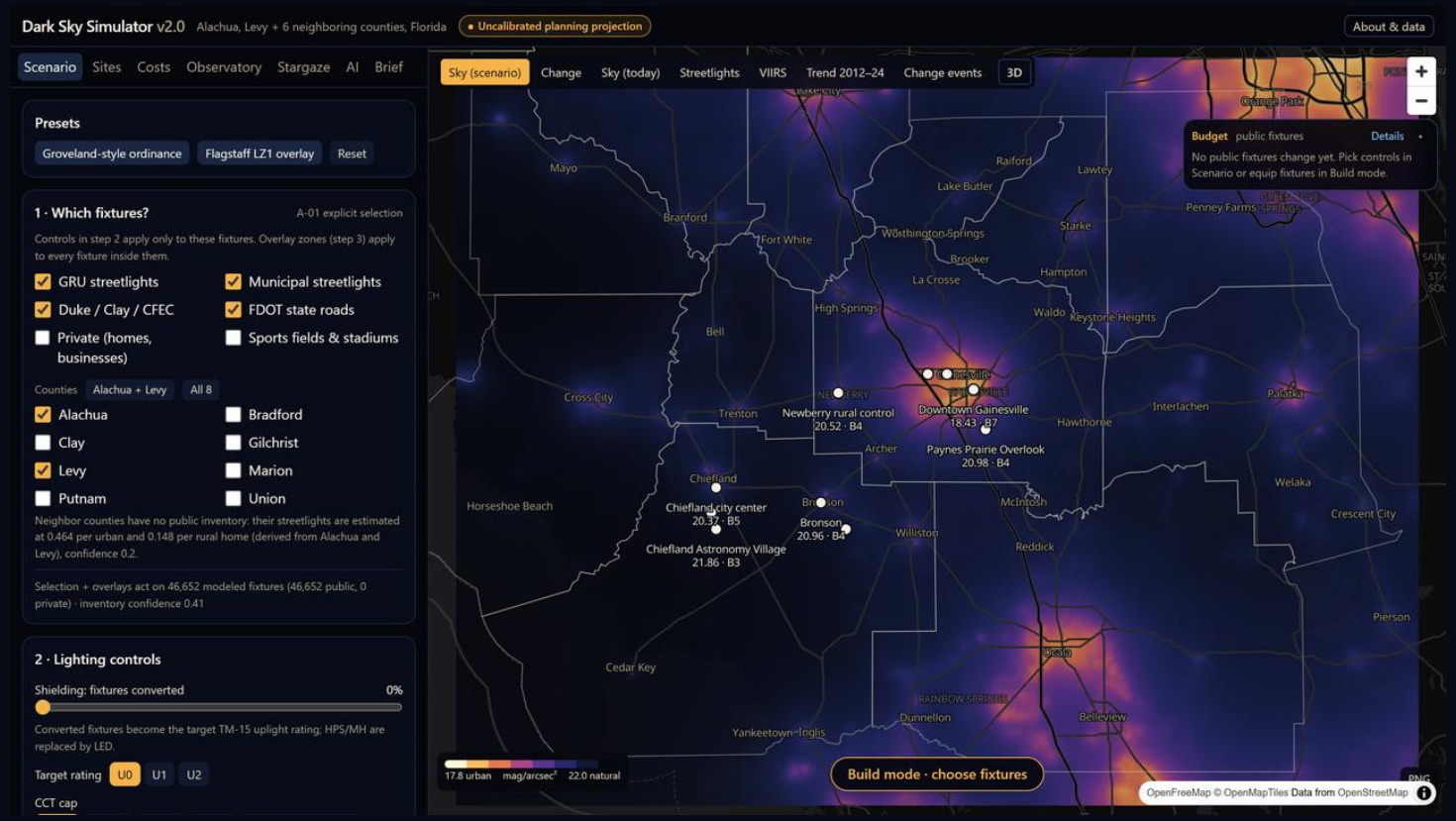
- Physics for the sky, public data for the lights, **AI only where data can teach**
- Every number labeled a planning projection; runs on a phone, no server

Sources: Rosen (2025), Dark Sky Simulator (v1) · Garstang (1986), Model for artificial night-sky illumination

THE SIMULATOR

# v2 at a glance

- **8 counties**: Alachua, Levy and six neighbors, ~0.4 km cells
- **~817,000** modeled light sources: streetlights, homes and businesses, sports fields
- Map: modeled zenith sky brightness (mag/arcsec$^2$); yellow = urban, black = near natural
- Seven tabs: Scenario · Sites · Costs · Observatory · Stargaze · **AI** · Brief



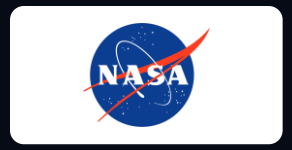

Scenario tab with today's modeled sky. Amber badge on every screen: “Uncalibrated planning projection”.

Sources: Román et al. (2018), NASA's Black Marble nighttime lights product suite · U.S. Census Bureau (2020), TIGER/Line roads and 2020 Census housing counts · OpenStreetMap contributors (2026), road, street-lamp and sports-venue data

HOW IT'S BUILT

# The technology stack

## Data




**NASA**
Black Marble




**NOAA**
VIIRS night lights




**USGS**
elevation, land cover




**U.S. Census**
housing, roads




**FEMA**
flood zones



**OpenStreetMap**
roads, lit streets

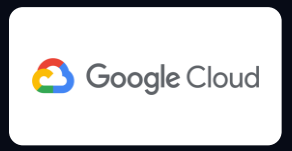



**Alachua County**
GIS, Policy 5.3.7

## Compute and ML



**Google Earth Engine**
satellite processing




**Google Cloud**
compute and scaling



**Python**
physics pipeline




**LightGBM**
growth forecast

## AI research and coding




**Gemini**
Deep Research



**Gemini Notebook**
organizing research

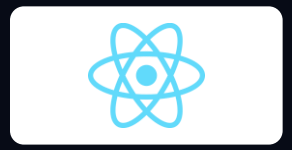



**Lovable (v1)**
AI app builder




**Claude Code (v2)**
AI coding

## App and hosting



**React**
web app




**TypeScript**
app code




**MapLibre**
maps



**Three.js**
3D streets



**Vite**
build



**GitHub**
code and tests



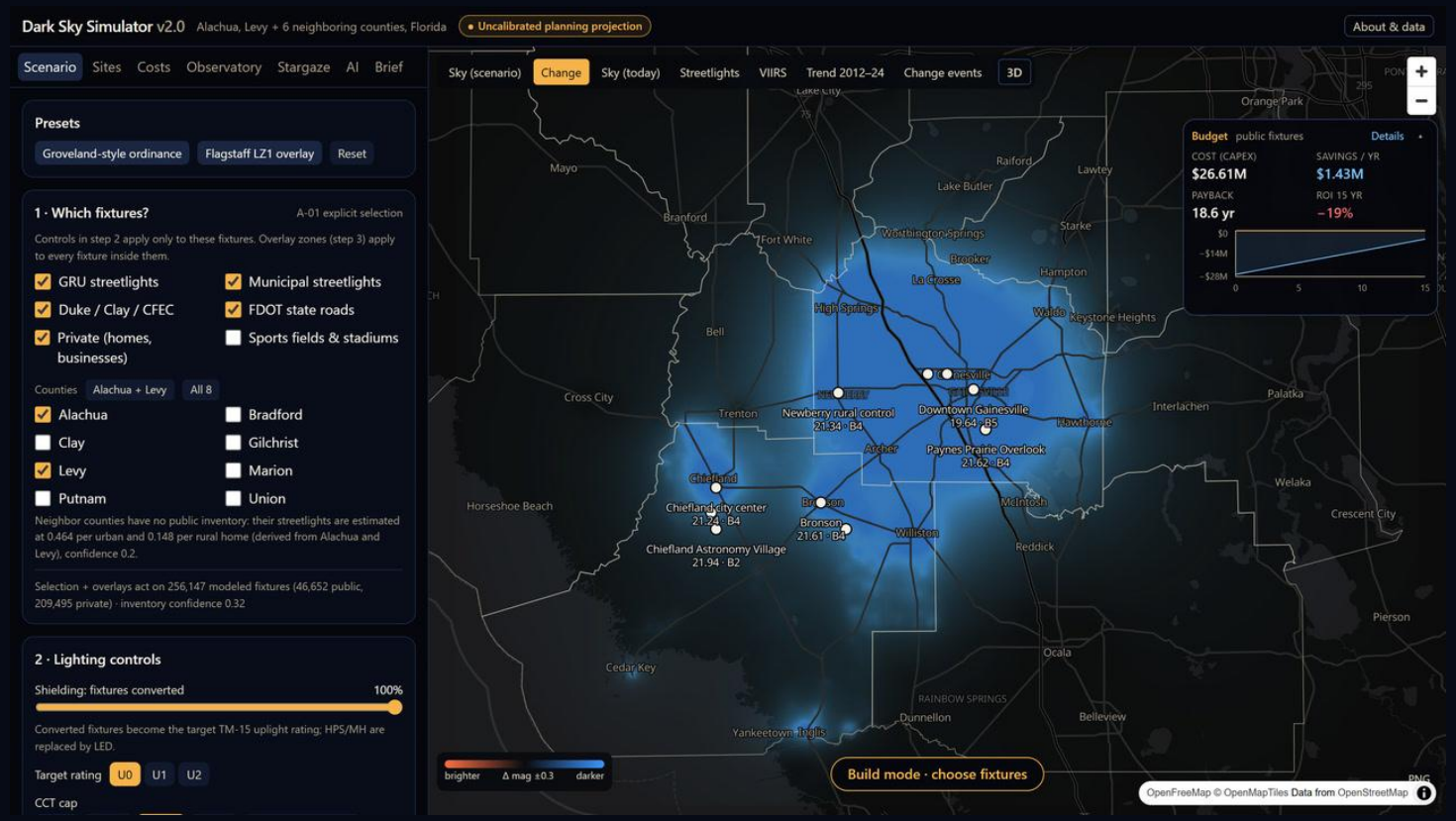

**Vercel**
hosting

Sources: Román et al. (2018), NASA's Black Marble nighttime lights product suite · Elvidge et al. (2021), Annual time series of global VIIRS nighttime lights · Dewitz (2023), National Land Cover Database (NLCD) 2021 · U.S. Census Bureau (2020), TIGER/Line roads and 2020 Census housing counts · FEMA (2026), National Flood Hazard Layer · OpenStreetMap contributors (2026), road, street-lamp and sports-venue data · Gorelick et al. (2017), Google Earth Engine: Planetary-scale geospatial analysis for everyone · Ke et al. (2017), LightGBM: A highly efficient gradient boosting decision tree

SCENARIO

# One click: a Groveland-style ordinance

- Groveland: Florida's first dark-sky city
- Full shielding (U0), **3000 K** cap, dim to 50% from 22:00 to 06:30
- Paynes Prairie: **20.98 → 21.62** mag/arcsec$^2$ if every light complies; **21.30** with public lights alone
- Blue = darker sky, and it crosses county lines
- Public cost: **$26.6M**, $1.43M/yr saved, 18.6-yr payback




Change layer: where the sky gets darker (blue) under the ordinance; Budget card at top right.

Sources: City of Groveland (2022), Ordinance 2022-26, outdoor lighting regulations · Garstang (1986), Model for artificial night-sky illumination · Bortle (2001), Introducing the Bortle dark-sky scale

SCENARIO · COSTS

# What each government pays, and saves

Right-sized public streetlights: **fully shielded 3000 K LEDs at 30% lower output**, no smart controllers. Savings = energy + avoided maintenance, at each utility's rates.

## Alachua County

all public streetlights (43,517)

| $15.3M | $2.55M | 6.0 yr | +$23.0M |
|---|---|---|---|
| one-time cost | saved / year | payback | after 15 years |

## University of Florida*

4,100 campus pole lights

| $984K | $198K | 5.0 yr | +$2.0M |
|---|---|---|---|
| one-time cost | saved / year | payback | after 15 years |

## City of Gainesville

GRU streetlights (30,000)

| $7.20M | $1.45M | 5.0 yr | +$14.6M |
|---|---|---|---|
| one-time cost | saved / year | payback | after 15 years |

## State roads (FDOT), Alachua

1,500 state-road lights

| $900K | $113K | 8.0 yr | +$795K |
|---|---|---|---|
| one-time cost | saved / year | payback | after 15 years |

## Levy County

all public streetlights (3,135)

| $1.88M | $228K | 8.3 yr | +$1.54M |
|---|---|---|---|
| one-time cost | saved / year | payback | after 15 years |

## City of Chiefland

285 city streetlights

| $171K | $21.5K | 8.0 yr | +$152K |
|---|---|---|---|
| one-time cost | saved / year | payback | after 15 years |

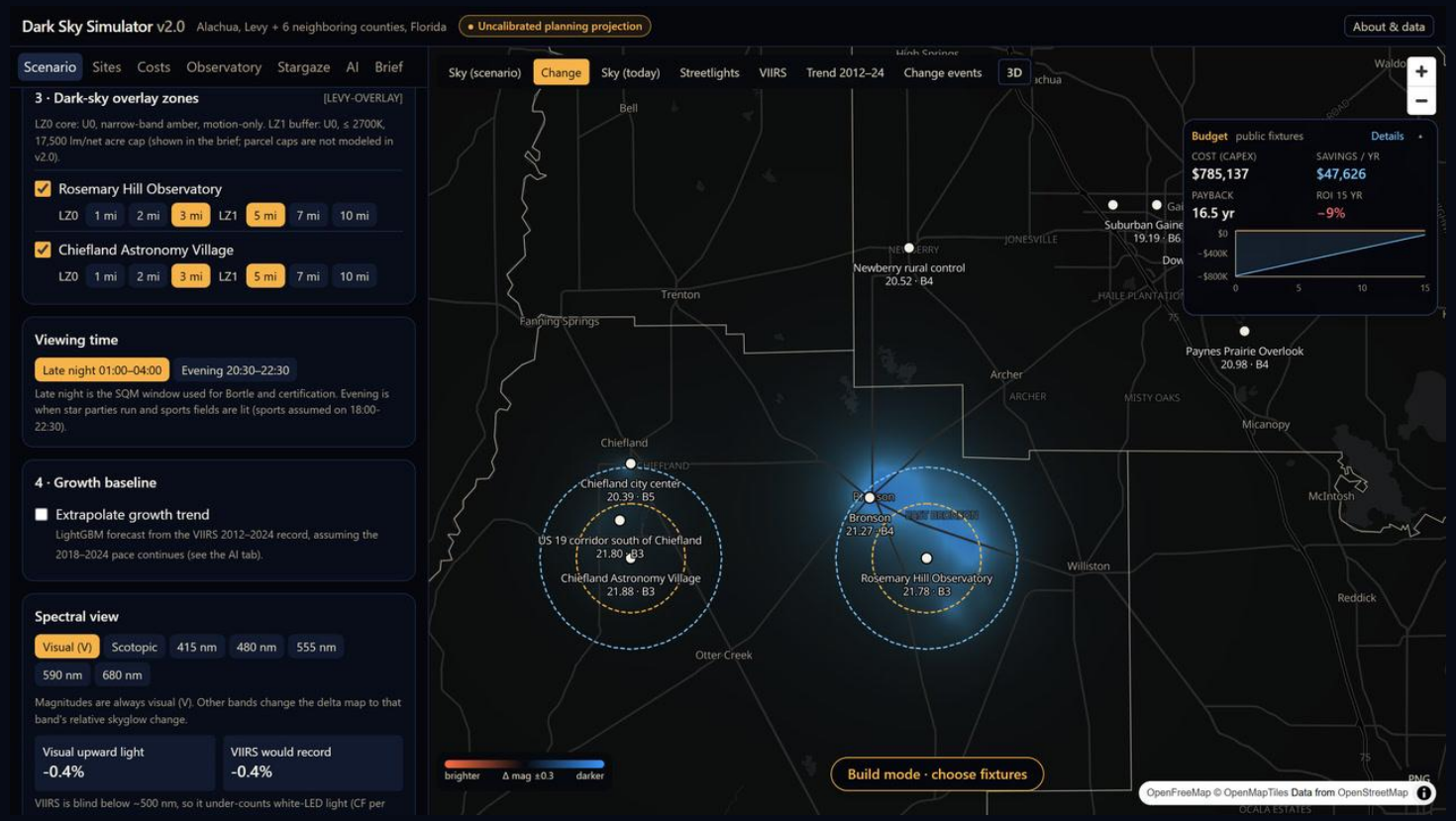
*UF: 4,100 campus pole lights, about 60% already LED (UF News, 2023), run through the simulator's cost engine at GRU rates. Gainesville and FDOT are within Alachua's total.

Sources: Dark Sky Simulator fixture catalog (2026) · Dark Sky Simulator research reports (2025), Gainesville Dark Sky Lighting Assessment and cost-benefit analyses · Roldán Hernández (2023), Security enhancements make for a brighter, safer campus (UF News)

SCENARIO

# Or protect just the observatories

- Overlay zones: stricter rules in rings around **Rosemary Hill** and **Chiefland Astronomy Village**
- LZ0 core: shielded, amber, motion-only
- LZ1 buffer: shielded, ≤ 2700 K
- Flagstaff-style preset: about **$0.79M** in public retrofits
- A trade-off a commission can actually vote on




Flagstaff LZ1 overlay preset (3-mile core, 5-mile buffer) on the Change layer.

Sources: Dark Sky Simulator fixture catalog (2026) · Dark Sky Simulator research reports (2025), Gainesville Dark Sky Lighting Assessment and cost-benefit analyses

SEE IT

# Paynes Prairie, today and with the ordinance




**Today**




**With a Groveland-style ordinance** (every light, public and private, complies)

All-sky view from Paynes Prairie Overlook, looking southeast as the Milky Way rises: May 28, 2022, from 1:30 AM, no Moon. Illustrative, not a photograph.

- **20.98** mag/arcsec$^2$ · Bortle 4
- Faintest naked-eye stars ≈ magnitude 6.1

- **21.62** mag/arcsec$^2$ (+0.64) · still Bortle 4
- Faintest stars ≈ 6.4: a clearer Milky Way

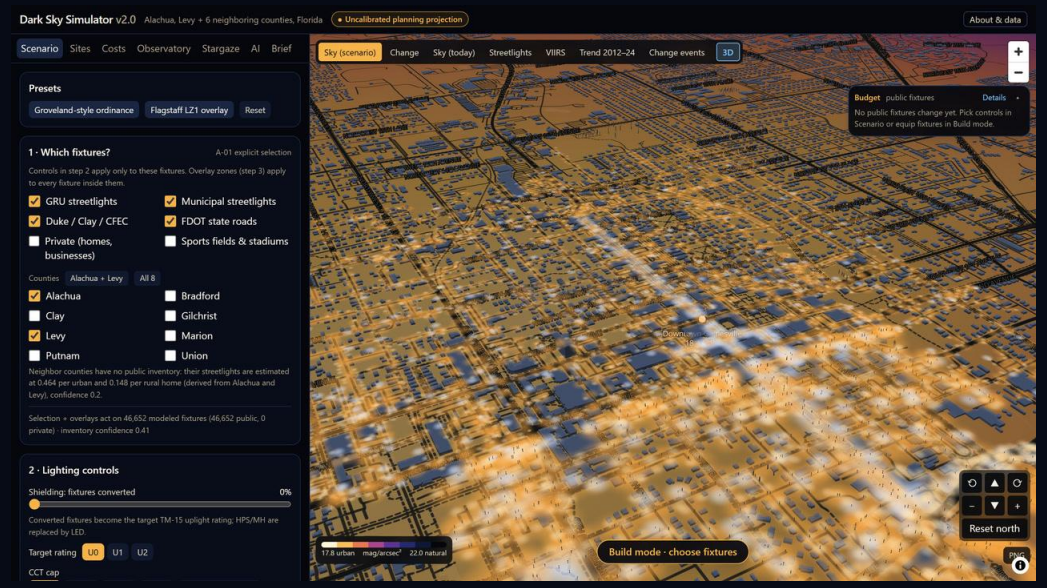
Sources: Garstang (1986), Model for artificial night-sky illumination · Bortle (2001), Introducing the Bortle dark-sky scale

SCENARIO

# Build mode and 3D: toward a digital twin




Build mode




3D: downtown Gainesville

- Catalog of **17 DarkSky-compliant fixture types**: spectrum, uplight rating, watts, installed cost
- Equip one: sky, budget, brief and install markers update
- Terrain, buildings and every modeled light at street level
- Mapped streetlights where they stand; modeled ones along the roads of their cell

**The goal: a digital twin** of the illumination and darkness of Gainesville and North Central Florida

Sources: DarkSky International, DarkSky Approved program · Dark Sky Simulator fixture catalog (2026) · OpenStreetMap contributors (2026), road, street-lamp and sports-venue data

# Where AI is used, and where it isn't

**1 · Data**
NASA Black Marble
13 years, 2012–2024

→

**2 · Cleaning**
remove the
solar-cycle airglow

→

**3 · Statistics**
step test on every
lit cell → change events

→

**4 · Machine learning**
LightGBM forecast
of where light grows

→ change-events map

→ 2034 sky and observatory site ranking

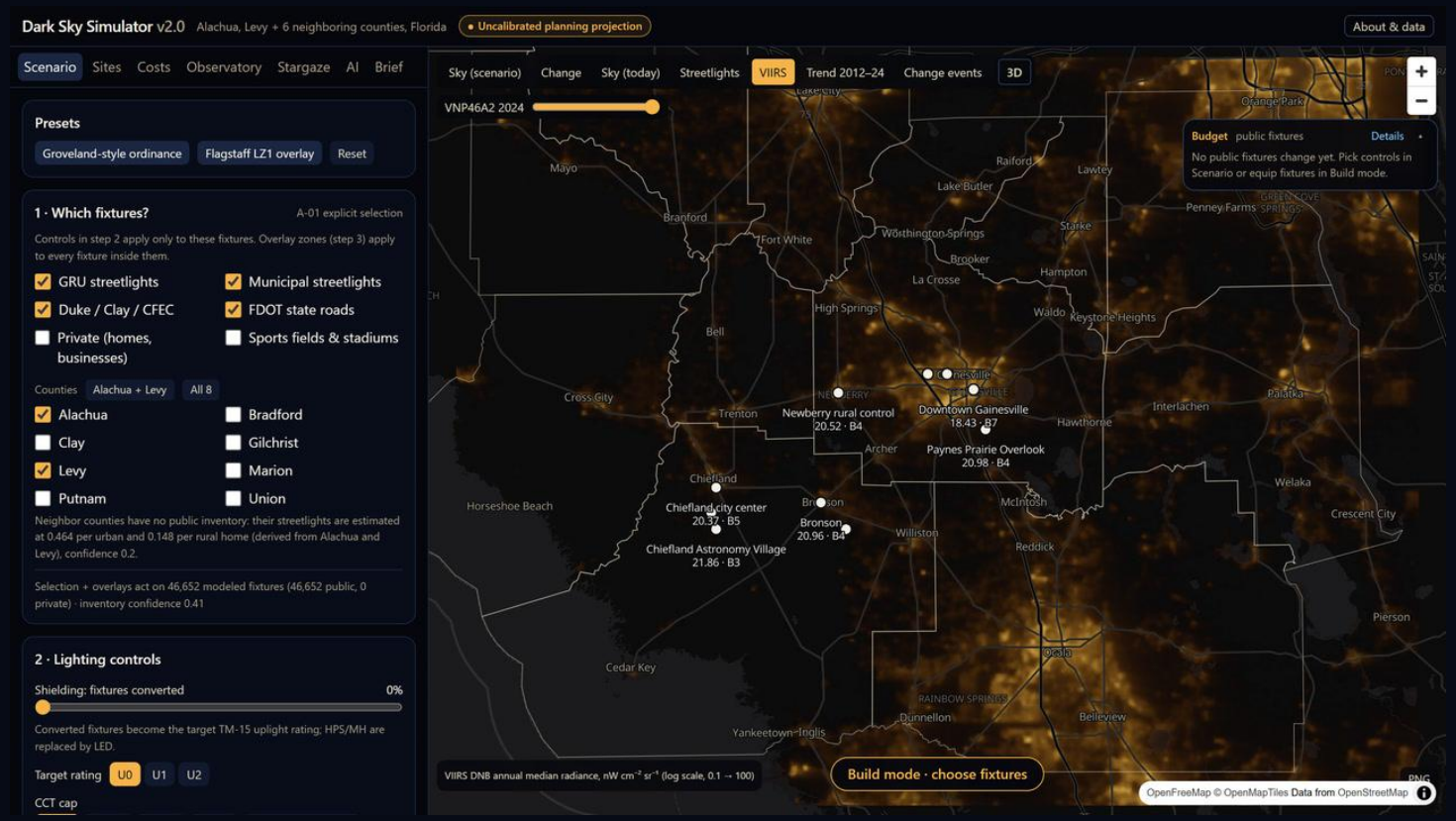
**NOT AI: physics and explicit rules you can inspect**

- **Sky brightness**: atmospheric scattering physics
- **Scenario effects**: published fixture rules
- **Site ranking**: weighted sum, weights you set
- **Costs**: unit prices × fixture counts

Sources: Román et al. (2018), NASA's Black Marble nighttime lights product suite · Andrews (1993), Tests for parameter instability and structural change with unknown change point · Ke et al. (2017), LightGBM: A highly efficient gradient boosting decision tree · Garstang (1986), Model for artificial night-sky illumination

# The data: 13 years of NASA night lights

- NASA Black Marble (**VIIRS VNP46A2**): nightly, moonlight- and angle-corrected, ~500 m
- Keep cloud-free, high-quality nights; **one median per year** per cell, 2012–2024
- Pulled through **Google Earth Engine**'s free noncommercial tier in ~12 minutes
- The 2024 layer also tells the physics model where private lighting is

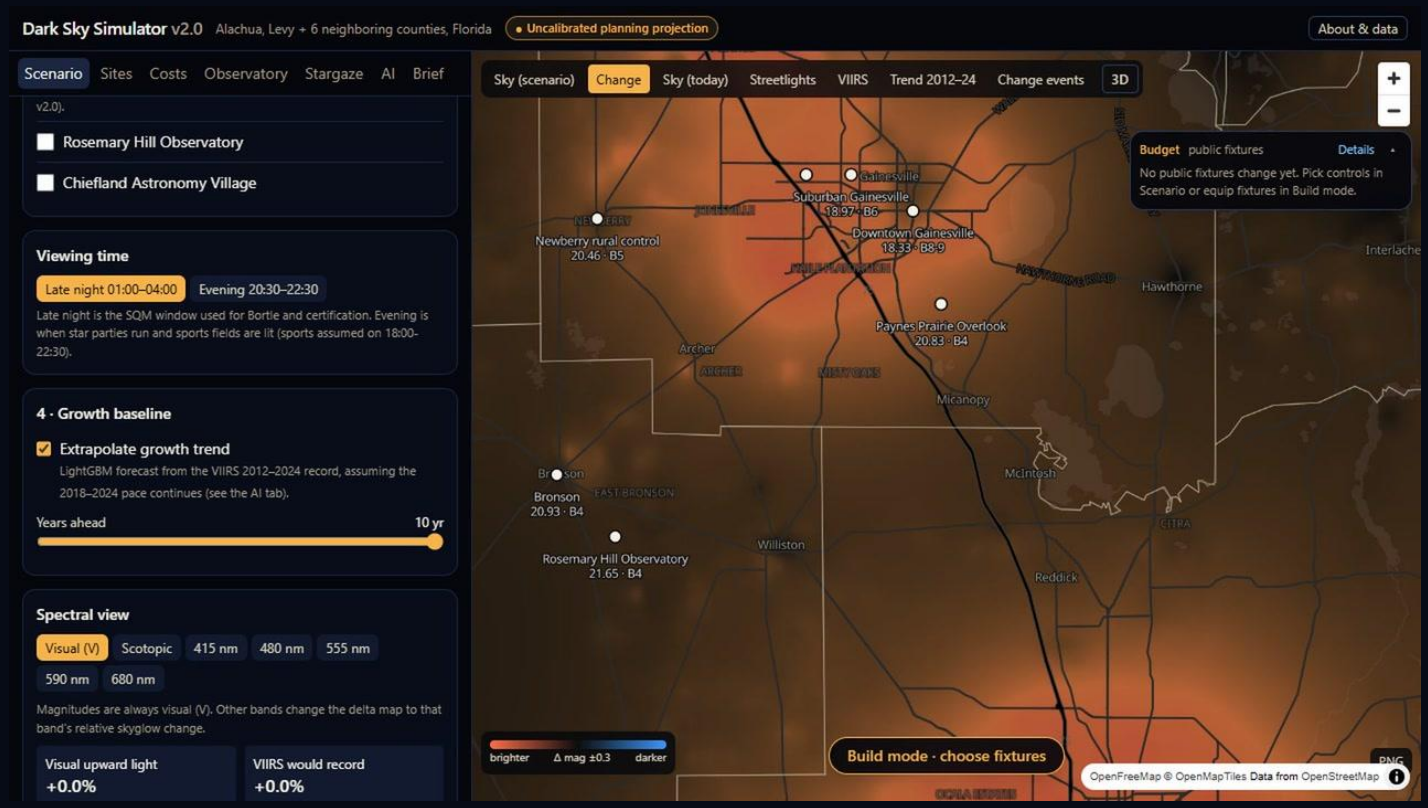



VIIRS layer: yearly median radiance with the airglow background removed; the slider steps through 2012–2024.

Sources: Román et al. (2018), NASA's Black Marble nighttime lights product suite · Gorelick et al. (2017), Google Earth Engine: Planetary-scale geospatial analysis for everyone

# What the AI found, and where the sky is heading

- **Found:** about 1,500 spots where light jumped suddenly since 2012, such as Celebration Pointe in 2016
- **Forecast:** LightGBM projects where light grows; the map steps the sky forward to **2034** if the 2018–24 pace continues
- **Cleaned first:** the upper atmosphere's natural glow follows the Sun's cycle; it's removed so the AI learns our lights, not the Sun
- **Checked:** the forecast is graded only on years it never saw



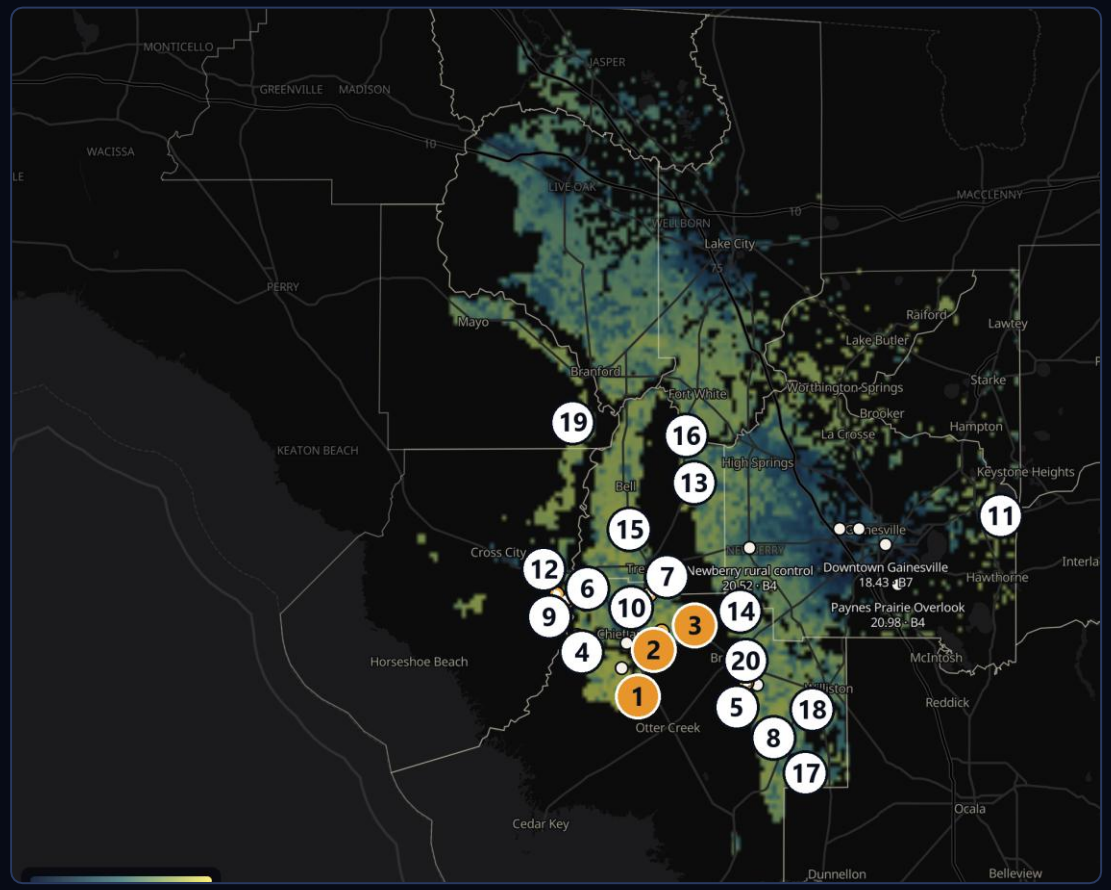

Scenario tab, growth forecast on: the sky stepping from 2025 to 2034 (orange = brighter than today).

Sources: Andrews (1993), Tests for parameter instability and structural change with unknown change point · Ke et al. (2017), LightGBM: A highly efficient gradient boosting decision tree · EMJ Construction (2016), Two new Florida Bass Pro locations underway · Román et al. (2018), NASA's Black Marble nighttime lights product suite

IN USE

# Where could UF build its next observatory?

- Requested by **Dr. Elizabeth Lada**, Chair of UF Astronomy
- Every ~1 km cell of 11 counties scored: dark sky today and in **2034** (LightGBM), clear air, land, access, **drive time from UF**
- Ruled out: **water and wetland**, the coast, ground below 5 m (storm surge), **2080 flood projections**, **FEMA's 100-year floodplain**, over ~90 min away
- **Shortlist: Levy #1–#3**, between Chiefland and Bronson: 44–58 min from campus, Bortle 3, land **$605K–806K** for 80 acres
- Telescope + building: $1–3M in published examples → **≈ $1.6–3.8M** total



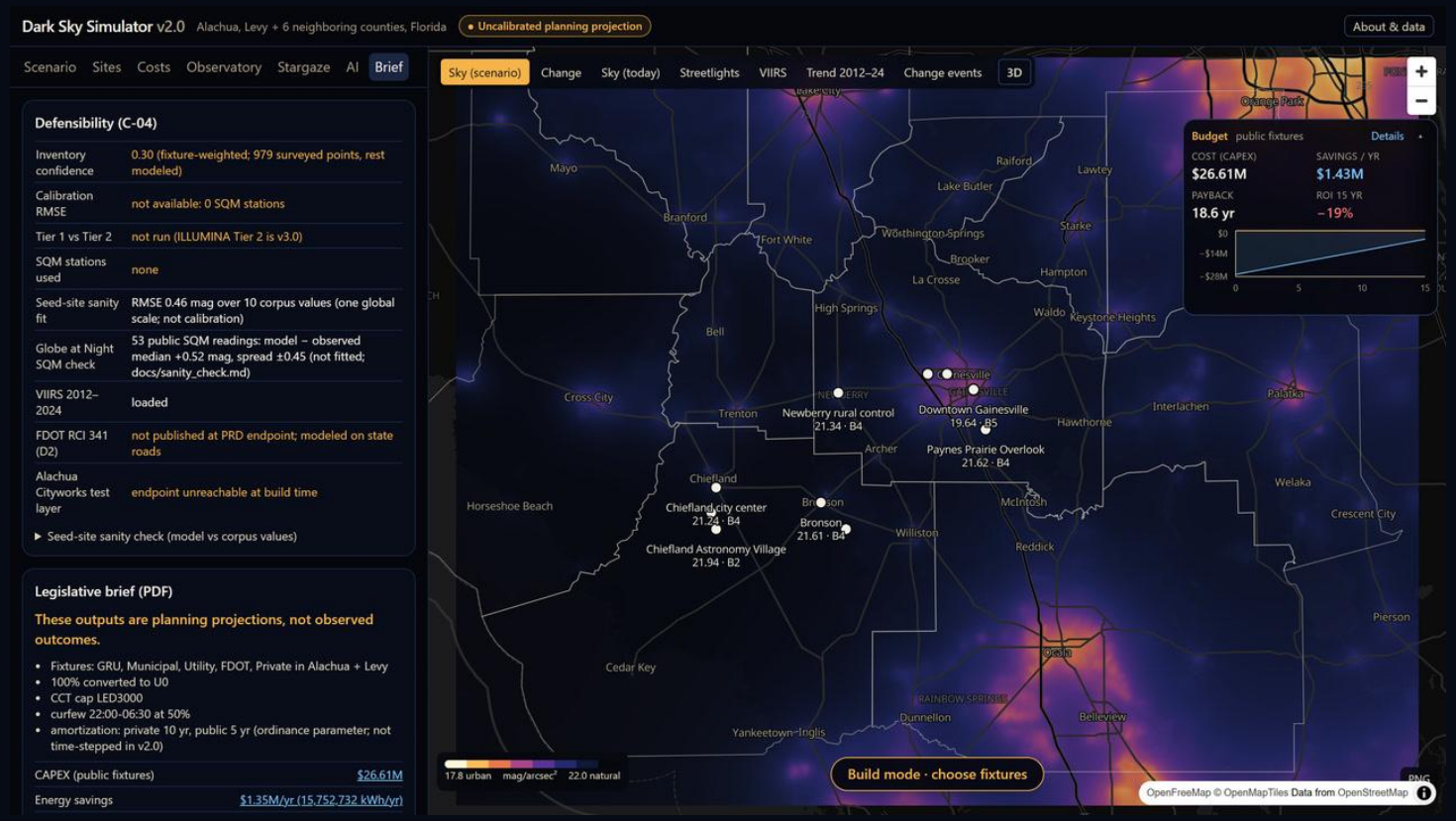

Observatory tab: the top 20 sites by rank (orange = Levy shortlist; markers enlarged). Blank areas are ruled out: water, wetland, flood-prone ground or over 90 minutes away.

Sources: Ke et al. (2017), LightGBM: A highly efficient gradient boosting decision tree · Dewitz (2023), National Land Cover Database (NLCD) 2021 · Cangialosi & Alaka (2026), Tropical Cyclone Report: Hurricane Idalia · Sweet et al. (2022), Global and Regional Sea Level Rise Scenarios for the United States · Ward et al. (2020), Aqueduct Floods Methodology · FEMA (2026), National Flood Hazard Layer · Florida Dept. of Revenue (2025), Tax roll and qualified sales · York University (2019), Installation of new telescope expected to be largest at a Canadian university · ABC7 Los Angeles (2011), Domes put in place for CSUSB's observatory · Florida Tech (2004), Funding secured for Florida Tech telescope

IN USE

# From model to policy brief

- One-click **PDF** for commissioners; every figure links back to its scenario
- First line: “These outputs are planning projections, not observed outcomes.”
- Defensibility panel lists what's missing: sky-meter stations, utility inventories, satellite calibration
- That list is the **v3 plan**

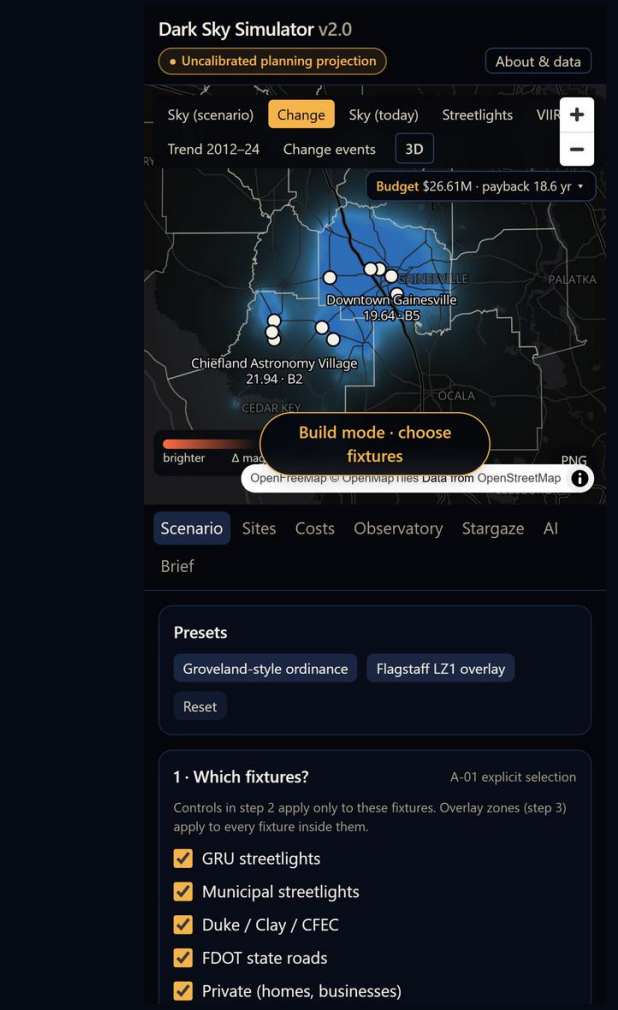



Brief tab: defensibility panel and the legislative brief.

WHAT'S NEXT

# Scaling up: from projection to evidence

- **Ground truth**: sky-quality meters in every county + a 50-node low-cost sensor pilot (~$5,900 hardware)
- **Real inventories**: utility data-sharing agreements (GRU, Duke, Clay, CFEC)
- **Computer vision**: a YOLO detector on street-level imagery to classify fixtures
- **Retrofit detection**: monthly satellite data; fit the LED correction on GRU's 3000 K conversions
- **Scaling**: a reusable county package (satellite data, lighting, development, policy), Levy County next
- **Siting**: parcel-level flood maps and site visits; blind test with UF Astronomy
- Looking for collaborators in sensors, computer vision, remote sensing and policy



The same app on a phone

WHAT'S NEXT

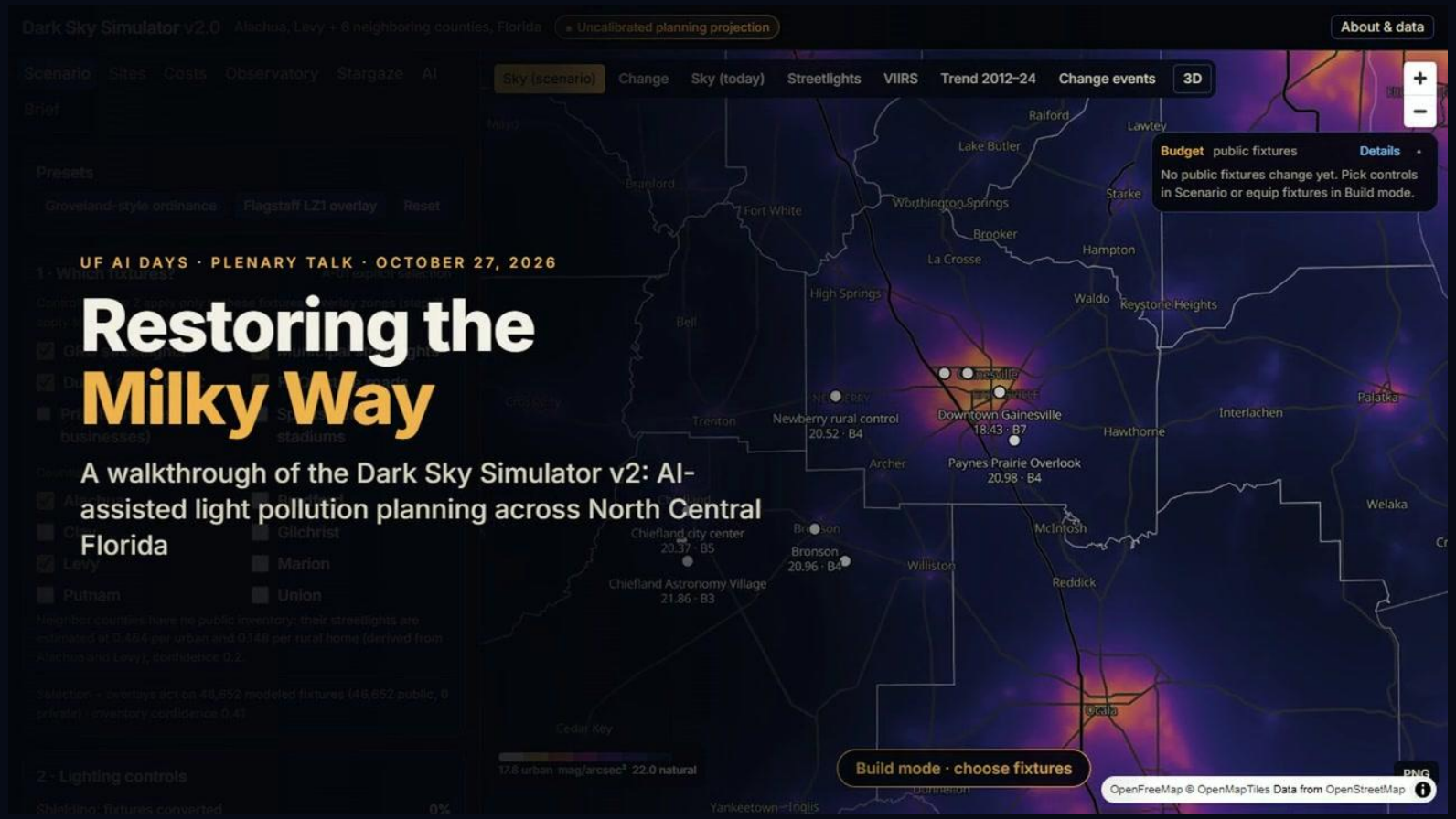
# What UF can do

- Alachua County has **adopted** dark-sky principles; **UF plans its own campus**, coordinated with the county, so it can lead
- **Lead on campus**: the model projects about **$984K** to upgrade UF's 4,100 pole lights, saving about **$198K** a year
- **Protect UF's skies**: keep Rosemary Hill dark and choose a site for UF Astronomy's next observatory
- **Measure the real sky**: a 50-sensor network (about **$6,000** in hardware) to check the model against real readings
- Independent and unfunded so far: I'd like this to become a UF project



UF's Rosemary Hill Observatory. Photo: The Gainesville Sun, April 2012.

Sources: Roldán Hernández (2023), Security enhancements make for a brighter, safer campus (UF News) · Dark Sky Simulator fixture catalog (2026)




# Thank you

## Questions?

**Try it**

dark-sky-simulator-v2.vercel.app/dark-sky

**Code**

github.com/jayrosen-design/dark-sky-simulator-v2

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